Subject: ASD&D | **Semester:** V

Class: TE | **Batch:** T11

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Experiment 2

Aim: To Containerize Application using Docker

Lab Outcome: Mapped LO4

LO4: Containerize, orchestrate and deploy applications using Docker, Docker Compose, Kubernetes and cloud infrastructure tools.

Theory:

- Docker

Docker is an open-source platform that enables developers to build, package, distribute, and run applications in lightweight, portable environments called containers. It includes everything an application needs to run, such as the application code, runtime, system libraries, and dependencies, ensuring consistent behavior across different environments. Docker operates using a client-server architecture.

- Containerization

Containerization is the process of packaging an application along with all of its dependencies into a single lightweight unit known as a container. Unlike traditional virtual machines (VMs) that require a full guest operating system, containers share the host operating system's kernel while maintaining isolated user spaces.

Key Advantages of Containerization:

➢ **Portability:** Runs consistently across various operating systems and cloud environments.

➢ **Speed:** Rapid application deployment and near-instant startup times.

➢ **Resource Efficiency:** Lower CPU, memory, and storage footprint compared to VMs.

- ➢ **Environment Consistency:** Eliminates the "works on my machine" problem.
- ➢ **Simplified Maintenance:** Streamlines application updates, scaling, and deployment workflows.

- Difference Between Docker Images and Containers

A Docker Image is a static, read-only template that contains the application source code, runtime environment, libraries, and system dependencies required to run an application. In contrast, a Docker Container is a dynamic, runnable, and isolated instance created directly from that image.

In terms of state, Docker images are immutable, meaning their underlying layers cannot be modified once they are built; any changes require creating a new image version. On the other hand, containers are mutable because Docker attaches a thin, writable layer on top of the image when instantiated, allowing files to be written or modified during runtime.

To use an analogy, a Docker Image acts as a blueprint, class, or installer executable, whereas a Docker Container represents the actual running process or object instantiated from that blueprint. Consequently, their lifecycles differ: images are built, pushed to or pulled from registries, stored locally, or deleted, while containers are actively managed through states such as created, started, stopped, restarted, and removed.

Commands:

docker --version

- ➢ **Purpose:** Checks and displays the installed version of the Docker client and daemon.

docker images

- ➢ **Purpose:** Lists all Docker images stored locally on the system along with their repository name, tag, image ID, and size.

docker search ubuntu

- ➢ **Purpose:** Searches Docker Hub for public images matching the term "ubuntu".

docker search ubuntu --filter stars=50

- **Purpose:** Searches Docker Hub for "ubuntu" images filtered to show only those with 50 or more stars (community ratings).

docker pull ubuntu

- **Purpose:** Downloads the official ubuntu:latest image from Docker Hub to the local host machine.

docker run ubuntu /bin/bash

- **Purpose:** Creates and runs a non-interactive container from the ubuntu image executing /bin/bash, which exits immediately because no terminal session is attached.

docker ps

- **Purpose:** Displays a list of all currently active (running) Docker containers.

docker run -it ubuntu /bin/bash

- **Purpose:** Launches an interactive container from the ubuntu image and opens a terminal (/bin/bash) session as the root user.

apt update

- **Purpose:** Updates the package lists and repository indexes inside the running Ubuntu container.

ls

- **Purpose:** Lists the files and directories in the current directory inside the container.

ls -la

- **Purpose:** Lists all files and directories inside the container, including hidden files, with detailed permissions and file attributes.

cd /tmp

- **Purpose:** Changes the current working directory to /tmp inside the container.

pwd

- ➢ **Purpose:** Prints the absolute path of the current working directory inside the container.

mkdir test

- ➢ **Purpose:** Creates a new directory named test inside /tmp.

cd test

- ➢ **Purpose:** Changes the working directory to the newly created test folder.

touch hello.txt

- ➢ **Purpose:** Creates an empty file named hello.txt inside the test directory.

echo "Hello Docker" > hello.txt

- ➢ **Purpose:** Writes the text string "Hello Docker" into hello.txt.

cat hello.txt

- ➢ **Purpose:** Reads and prints the contents of hello.txt to the terminal screen.

cat /etc/os-release

- ➢ **Purpose:** Displays the operating system distribution details and version information of the running container.

whoami

- ➢ **Purpose:** Displays the username of the currently logged-in user inside the container (root).

echo "Docker is awesome!" > myfile.txt

- ➢ **Purpose:** Creates a file named myfile.txt and writes the string "Docker is awesome!" into it.

cat myfile.txt

- **Purpose:** Reads and prints the contents of myfile.txt to the terminal screen.

exit

- **Purpose:** Terminates the interactive bash session and exits the running container.

docker ps -a

- **Purpose:** Lists all containers on the system, including both running and exited/stopped containers.

docker pull nginx

- **Purpose:** Downloads the official nginx web server image from Docker Hub.

docker search redis

- **Purpose:** Searches Docker Hub for public images matching the term "redis".

docker rmi ubuntu

- **Purpose:** Attempts to remove the local ubuntu image from the host system.

docker run -it ubuntu

- **Purpose:** Launches a new interactive Ubuntu container with default terminal settings.

docker run -d --name webserver -p 8081:80 nginx

- **Purpose:** Runs an NGINX container in detached (background) mode named webserver, mapping host port 8081 to container port 80.

docker logs webserver

- **Purpose:** Retrieves and displays the runtime standard output and error logs produced by the webserver container.

docker stop webserver

- **Purpose:** Gracefully stops the running webserver container.

docker start webserver

➢ **Purpose:** Starts the stopped webserver container.

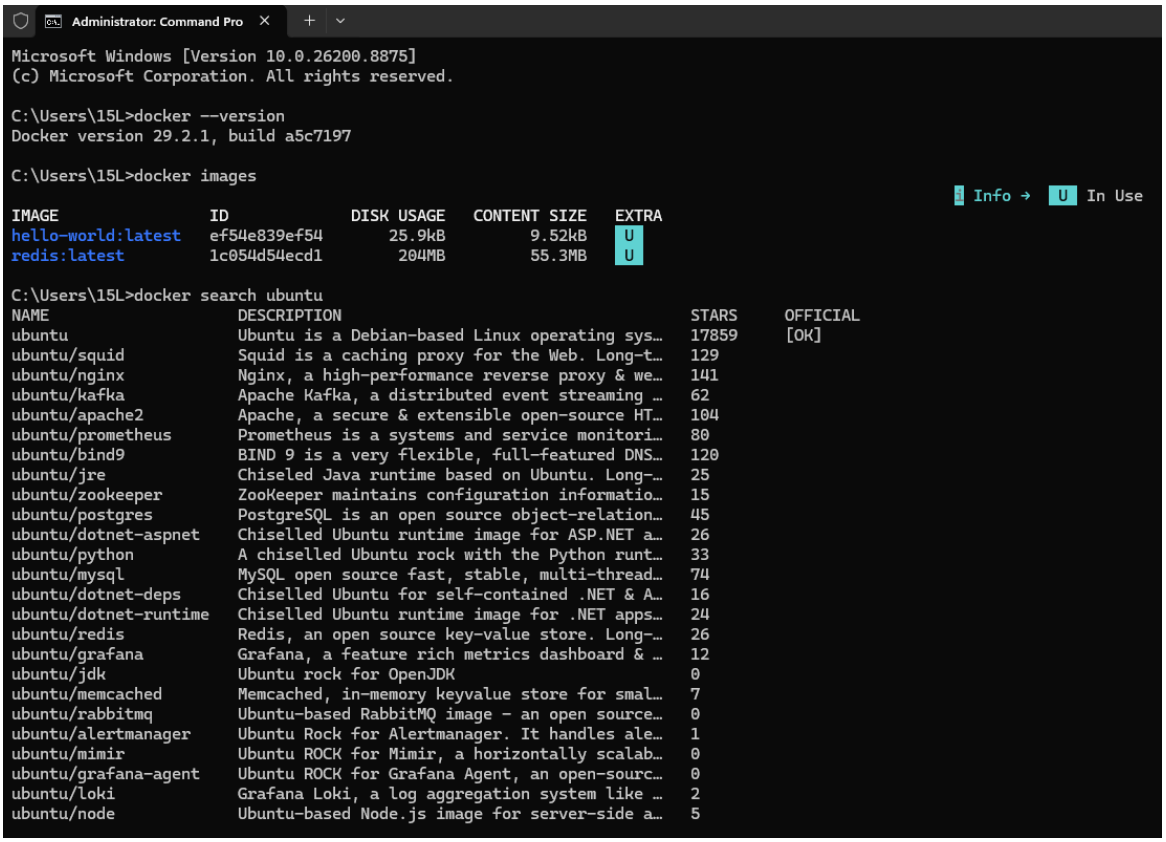
docker restart webserver

➢ **Purpose:** Restarts the running webserver container.

docker inspect webserver

➢ **Purpose:** Generates a detailed JSON array containing low-level operational and configuration data for the webserver container (such as IP address, status, and port mappings).

Output:



```
C:\Users\15L>docker search ubuntu --filter stars=50
NAME                DESCRIPTION                                     STARS     OFFICIAL
ubuntu              Ubuntu is a Debian-based Linux operating sys…   17859     [OK]
ubuntu/squid        Squid is a caching proxy for the Web. Long-t…   129
ubuntu/nginx        Nginx, a high-performance reverse proxy & we…   141
ubuntu/kafka        Apache Kafka, a distributed event streaming …   62
ubuntu/bind9        BIND 9 is a very flexible, full-featured DNS…   120
ubuntu/apache2      Apache, a secure & extensible open-source HT…   104
ubuntu/prometheus   Prometheus is a systems and service monitori…   80
ubuntu/mysql        MySQL open source fast, stable, multi-thread…   74
```

```
C:\Users\15L>docker pull ubuntu
Using default tag: latest
latest: Pulling from library/ubuntu
a3679419df18: Pull complete
ed819469700f: Pull complete
e16351a257e4: Download complete
Digest: sha256:3131b4cc82a783df6c9df078f86e01819a13594b865c2cad47bd1bca2b7063bb
Status: Downloaded newer image for ubuntu:latest
docker.io/library/ubuntu:latest

C:\Users\15L>docker run ubuntu /bin/bash

C:\Users\15L>docker ps
CONTAINER ID   IMAGE     COMMAND   CREATED   STATUS    PORTS     NAMES

C:\Users\15L>docker run -it ubuntu /bin/bash
root@b00193fd8c8f:/# apt update
Get:1 http://security.ubuntu.com/ubuntu resolute-security InRelease [137 kB]
Get:2 http://archive.ubuntu.com/ubuntu resolute InRelease [136 kB]
Get:3 http://security.ubuntu.com/ubuntu resolute-security/universe amd64 Packages [170 kB]
Get:4 http://archive.ubuntu.com/ubuntu resolute-updates InRelease [137 kB]
Get:5 http://archive.ubuntu.com/ubuntu resolute-backports InRelease [136 kB]
Get:6 http://security.ubuntu.com/ubuntu resolute-security/restricted amd64 Packages [303 kB]
Get:7 http://archive.ubuntu.com/ubuntu resolute/main amd64 Packages [1874 kB]
Get:8 http://security.ubuntu.com/ubuntu resolute-security/main amd64 Packages [361 kB]
Get:9 http://archive.ubuntu.com/ubuntu resolute/multiverse amd64 Packages [352 kB]
Get:10 http://archive.ubuntu.com/ubuntu resolute/restricted amd64 Packages [189 kB]
Get:11 http://archive.ubuntu.com/ubuntu resolute/universe amd64 Packages [20.1 MB]
Get:12 http://archive.ubuntu.com/ubuntu resolute-updates/multiverse amd64 Packages [3584 B]
Get:13 http://archive.ubuntu.com/ubuntu resolute-updates/restricted amd64 Packages [316 kB]
Get:14 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 Packages [442 kB]
Get:15 http://archive.ubuntu.com/ubuntu resolute-updates/universe amd64 Packages [261 kB]
Fetched 24.9 MB in 6s (4425 kB/s)
1 package can be upgraded. Run 'apt list --upgradable' to see it.
```

```
root@b00193fd8c8f:/# git -version
unknown option: -version
usage: git [-v | --version] [-h | --help] [-C <path>] [-c <name>=<value>]
           [--exec-path[=<path>]] [--html-path] [--man-path] [--info-path]
           [-p | --paginate | -P | --no-pager] [--no-replace-objects] [--no-lazy-fetch]
           [--no-optional-locks] [--no-advice] [--bare] [--git-dir=<path>]
           [--work-tree=<path>] [--namespace=<name>] [--config-env=<name>=<envvar>]
           <command> [<args>]
root@b00193fd8c8f:/# curl --version
curl 8.18.0 (x86_64-pc-linux-gnu) libcurl/8.18.0 OpenSSL/3.5.5 zlib/1.3.1 brotli/1.2.0 zstd/1.5.7 libidn2/2.3.8 libpsl/0.21.2 libssh2/1.11.1 nghttp2/1.68.0 librtmp/2.3 mit-krb5/1.22.1 OpenLDAP/2.6.10
Release-Date: 2026-01-07, security patched: 8.18.0-1ubuntu2.3
Protocols: dict file ftp ftps gopher gophers http https imap imaps ipfs ipns ldap ldaps mqtt pop3 pop3s rtmp rtsp scp sftp smb smbs smtp smtps telnet tftp ws wss
Features: alt-svc AsynchDNS brotli GSS-API HSTS HTTP2 HTTPS-proxy IDN IPv6 Kerberos Largefile libz NTLM PSL SPNEGO SSL threadsafe TLS-SRP UnixSockets zstd
root@b00193fd8c8f:/# ls
bin  boot  dev  etc  home  lib  lib64  media  mnt  opt  proc  root  run  sbin  srv  sys  tmp  usr  var
root@b00193fd8c8f:/# ls -la
total 72
drwxr-xr-x   1 root root 4096 Jul 20 13:05 .
drwxr-xr-x   1 root root 4096 Jul 20 13:05 ..
-rwxr-xr-x   1 root root    0 Jul 20 13:05 .dockerenv
drwxr-xr-x   2 root root 4096 Jul 13 19:06 .rock
lrwxrwxrwx   1 root root    7 Apr 20 11:46 bin -> usr/bin
drwxr-xr-x   2 root root 4096 Apr 20 11:46 boot
drwxr-xr-x   5 root root  360 Jul 20 13:05 dev
drwxr-xr-x   1 root root 4096 Jul 20 13:18 etc
drwxr-xr-x   3 root root 4096 Jul 13 19:06 home
lrwxrwxrwx   1 root root    7 Apr 20 11:46 lib -> usr/lib
lrwxrwxrwx   1 root root    9 Apr 20 11:46 lib64 -> usr/lib64
drwxr-xr-x   2 root root 4096 Jul 13 19:05 media
drwxr-xr-x   2 root root 4096 Jul 13 19:05 mnt
drwxr-xr-x   2 root root 4096 Jul 13 19:05 opt
dr-xr-xr-x 348 root root    0 Jul 20 13:05 proc
drwx------   2 root root 4096 Jul 13 19:06 root
drwxr-xr-x   1 root root 4096 Jul 20 13:18 run
lrwxrwxrwx   1 root root    8 Apr 20 11:46 sbin -> usr/sbin
drwxr-xr-x   2 root root 4096 Jul 13 19:05 srv
dr-xr-xr-x  13 root root    0 Jul 20 13:05 sys
drwxrwxrwt   1 root root 4096 Jul 20 13:18 tmp
drwxr-xr-x   1 root root 4096 Jul 13 19:05 usr
drwxr-xr-x   1 root root 4096 Jul 13 19:06 var
root@b00193fd8c8f:/# cd /tmp
root@b00193fd8c8f:/tmp# pwd
/tmp
root@b00193fd8c8f:/tmp# mkdir test
root@b00193fd8c8f:/tmp# cd test
root@b00193fd8c8f:/tmp/test# touch hello.txt
root@b00193fd8c8f:/tmp/test# cho "Hello Docker" > hello.txt
```

```
root@b00193fd8c8f:/# cd /tmp
root@b00193fd8c8f:/tmp# pwd
/tmp
root@b00193fd8c8f:/tmp# mkdir test
root@b00193fd8c8f:/tmp# cd test
root@b00193fd8c8f:/tmp/test# touch hello.txt
root@b00193fd8c8f:/tmp/test# cho "Hello Docker" > hello.txt
bash: cho: command not found
root@b00193fd8c8f:/tmp/test# echo "Hello Docker" > hello.txt
root@b00193fd8c8f:/tmp/test# cat hello.txt
Hello Docker
root@b00193fd8c8f:/tmp/test# cat /etc/os-release
PRETTY_NAME="Ubuntu 26.04 LTS"
NAME="Ubuntu"
VERSION_ID="26.04"
VERSION="26.04 LTS (Resolute Raccoon)"
VERSION_CODENAME=resolute
ID=ubuntu
ID_LIKE=debian
HOME_URL="https://www.ubuntu.com/"
SUPPORT_URL="https://help.ubuntu.com/"
BUG_REPORT_URL="https://bugs.launchpad.net/ubuntu/"
PRIVACY_POLICY_URL="https://www.ubuntu.com/legal/terms-and-policies/privacy-policy"
UBUNTU_CODENAME=resolute
LOGO=ubuntu-logo
root@b00193fd8c8f:/tmp/test# whoami
root
root@b00193fd8c8f:/tmp/test# echo "Docker is awesome!" > myfile.txt
root@b00193fd8c8f:/tmp/test# cat myfile.txt
Docker is awesome!
root@b00193fd8c8f:/tmp/test# exit
exit

C:\Users\15L>dockerps -a
'dockerps' is not recognized as an internal or external command,
operable program or batch file.

C:\Users\15L>docker ps -a
CONTAINER ID   IMAGE          COMMAND                  CREATED          STATUS                      PORTS      NAMES
b00193fd8c8f   ubuntu         "/bin/bash"              18 minutes ago   Exited (0) 17 seconds ago              jolly_hofstadter
ce280af76907   ubuntu         "/bin/bash"              21 minutes ago   Exited (0) 21 minutes ago              sharp_pasteur
04ff3ca987f1   hello-world    "/hello"                 4 months ago     Exited (0) 4 months ago                focused_agnesi
506fbcc93880   redis:latest   "docker-entrypoint.s…"   4 months ago     Exited (255) 32 minutes ago   6379/tcp   distracted_babbage
d3e8d53b8e82   redis          "docker-entrypoint.s…"   4 months ago     Exited (255) 32 minutes ago   6379/tcp   competent_rosalind
```

```
C:\Users\15L>docker images
                                                                    i Info →   U  In Use
IMAGE                ID             DISK USAGE   CONTENT SIZE   EXTRA
hello-world:latest   ef54e839ef54       25.9kB         9.52kB   U
nginx:latest         5a88c9c45479        241MB           66MB
redis:latest         1c054d54ecd1        204MB         55.3MB   U
ubuntu:latest        3131b4cc82a7        161MB         45.3MB   U

C:\Users\15L>docker pull ubuntu
Using default tag: latest
latest: Pulling from library/ubuntu
Digest: sha256:3131b4cc82a783df6c9df078f86e01819a13594b865c2cad47bd1bca2b7063bb
Status: Image is up to date for ubuntu:latest
docker.io/library/ubuntu:latest

C:\Users\15L>docker pull nginx
Using default tag: latest
latest: Pulling from library/nginx
Digest: sha256:5a88c9c45479443d7be2eadc894b4ed0a9801bae03d97a5760ae13b5c2005942
Status: Image is up to date for nginx:latest
docker.io/library/nginx:latest

C:\Users\15L>docker search redis
NAME                                DESCRIPTION                                     STARS     OFFICIAL
redis                               Redis is the world's fastest data platform f…   13604     [OK]
redis/redis-stack-server            redis-stack-server installs a Redis server w…   104
redis/redis-stack                   redis-stack installs a Redis server with add…   162
redis/redisinsight                  Redis Insight - our best official GUI for Re…   54
redislabs/redis                     Clustered in-memory database engine compatib…   46
redis/rdi-api                                                                       0
redis/rdi-operator                                                                  0
redis/rdi-monitor                                                                   0
redis/rdi-collector-initializer      Init container for RDI Collector               0
redis/rdi-collector-api                                                             0
redis/rdi-processor                                                                 1
redis/rdi-cli                                                                       0
redis/rdi-metrics-aggregator                                                        0
redis/rdi-flink-processor                                                           0
redis/rdi-flink-collector                                                           0
redis/reloader                                                                      0
redis/sidekiq-benchmark             Sidekiq load benchmark in Rust - throughput …   0
circleci/redis                      CircleCI images for Redis                       18
redis/redis-benchmark-go            redis-benchmark utility in go.                  1
redis/vector-db-benchmark           Framework for benchmarking vector search eng…   1
bitnamicharts/redis                 Bitnami Helm chart for Redis(R)                 37
redis/bcast-tracking-bench          Redis CLIENT TRACKING BCAST invalidation lis…   0
redis/redis-di-cli                                                                  0
redis/pubsub-sub-bench                                                              0
cimg/redis                                                                          2
```

```
C:\Users\15L>docker rmi ubuntu
Error response from daemon: conflict: unable to delete ubuntu:latest (must be forced) - container ce280af76907 is using its referenced image 3131b4cc82a7

C:\Users\15L>docker run -it ubuntu
root@1000bc0dcc79:/# ls
bin  boot  dev  etc  home  lib  lib64  media  mnt  opt  proc  root  run  sbin  srv  sys  tmp  usr  var
root@1000bc0dcc79:/# pwd
/
root@1000bc0dcc79:/# whoami
root
root@1000bc0dcc79:/# exit
exit
```

```
C:\Users\15L>docker run -d --name webserver -p 8081:80 nginx
e637cbaba0f0bd0fc54fb656a56642dfbdd32e9c382a2a86a1b4d82fb372a5c5

C:\Users\15L>docker ps
CONTAINER ID   IMAGE     COMMAND                  CREATED          STATUS          PORTS                                     NAMES
e637cbaba0f0   nginx     "/docker-entrypoint.…"   47 seconds ago   Up 43 seconds   0.0.0.0:8081->80/tcp, [::]:8081->80/tcp   webserver
e41f9ed60c3c   nginx     "/docker-entrypoint.…"   7 minutes ago    Up 4 minutes    80/tcp                                    mynginx

C:\Users\15L>docker ps -a
CONTAINER ID   IMAGE          COMMAND                  CREATED          STATUS                      PORTS                                     NAMES
e637cbaba0f0   nginx          "/docker-entrypoint.…"   56 seconds ago   Up 49 seconds               0.0.0.0:8081->80/tcp, [::]:8081->80/tcp   webserver
f7b09cfcb56e   nginx          "/docker-entrypoint.…"   3 minutes ago    Created                                                       practice
e41f9ed60c3c   nginx          "/docker-entrypoint.…"   7 minutes ago    Up 4 minutes                80/tcp                                    mynginx
1000bc0dcc79   ubuntu         "/bin/bash"              9 minutes ago    Exited (0) 8 minutes ago                                      mystifying_wilson
b00193fd8c8f   ubuntu         "/bin/bash"              38 minutes ago   Exited (0) 18 minutes ago                                     jolly_hofstadter
ce280af76907   ubuntu         "/bin/bash"              41 minutes ago   Exited (0) 41 minutes ago                                     sharp_pasteur
04ff3ca987f1   hello-world    "/hello"                 4 months ago     Exited (0) 4 months ago                                       focused_agnesi
506fbcc93880   redis:latest   "docker-entrypoint.s…"   4 months ago     Exited (255) 52 minutes ago   6379/tcp                        distracted_babbage
d3e8d53b8e82   redis          "docker-entrypoint.s…"   4 months ago     Exited (255) 52 minutes ago   6379/tcp                        competent_rosalind

C:\Users\15L>docker logs webserver
/docker-entrypoint.sh: /docker-entrypoint.d/ is not empty, will attempt to perform configuration
/docker-entrypoint.sh: Looking for shell scripts in /docker-entrypoint.d/
/docker-entrypoint.sh: Launching /docker-entrypoint.d/10-listen-on-ipv6-by-default.sh
10-listen-on-ipv6-by-default.sh: info: Getting the checksum of /etc/nginx/conf.d/default.conf
10-listen-on-ipv6-by-default.sh: info: Enabled listen on IPv6 in /etc/nginx/conf.d/default.conf
/docker-entrypoint.sh: Sourcing /docker-entrypoint.d/15-local-resolvers.envsh
/docker-entrypoint.sh: Launching /docker-entrypoint.d/20-envsubst-on-templates.sh
/docker-entrypoint.sh: Launching /docker-entrypoint.d/30-tune-worker-processes.sh
/docker-entrypoint.sh: Configuration complete; ready for start up
2026/07/20 10:43:04 [notice] 1#1: using the "epoll" event method
2026/07/20 10:43:04 [notice] 1#1: nginx/1.31.3
2026/07/20 10:43:04 [notice] 1#1: built by gcc 14.2.0 (Debian 14.2.0-19)
2026/07/20 10:43:04 [notice] 1#1: OS: Linux 6.18.33.2-microsoft-standard-WSL2
2026/07/20 10:43:04 [notice] 1#1: getrlimit(RLIMIT_NOFILE): 1048576:1048576
2026/07/20 10:43:04 [notice] 1#1: start worker processes
2026/07/20 10:43:04 [notice] 1#1: start worker process 29
2026/07/20 10:43:04 [notice] 1#1: start worker process 30
2026/07/20 10:43:04 [notice] 1#1: start worker process 31
2026/07/20 10:43:04 [notice] 1#1: start worker process 32
2026/07/20 10:43:04 [notice] 1#1: start worker process 33
2026/07/20 10:43:04 [notice] 1#1: start worker process 34
2026/07/20 10:43:04 [notice] 1#1: start worker process 35
2026/07/20 10:43:04 [notice] 1#1: start worker process 36
2026/07/20 10:43:04 [notice] 1#1: start worker process 37
2026/07/20 10:43:04 [notice] 1#1: start worker process 38
2026/07/20 10:43:04 [notice] 1#1: start worker process 39
2026/07/20 10:43:04 [notice] 1#1: start worker process 40
2026/07/20 10:43:04 [notice] 1#1: start worker process 41
```

```
C:\Users\15L>docker stop webserver
webserver

C:\Users\15L>docker start webserver
webserver

C:\Users\15L>docker restart webserver
webserver

C:\Users\15L>docker inspect webserver
[
    {
        "Id": "e637cbaba0f0bd0fc54fb656a56642dfbdd32e9c382a2a86a1b4d82fb372a5c5",
        "Created": "2026-07-20T10:43:04.689271852Z",
        "Path": "/docker-entrypoint.sh",
        "Args": [
            "nginx",
            "-g",
            "daemon off;"
        ],
        "State": {
            "Status": "running",
            "Running": true,
            "Paused": false,
            "Restarting": false,
            "OOMKilled": false,
            "Dead": false,
            "Pid": 4161,
            "ExitCode": 0,
            "Error": "",
            "StartedAt": "2026-07-20T10:44:55.2247221Z",
            "FinishedAt": "2026-07-20T10:44:54.9997411286Z"
        },
        "Image": "sha256:5a88c9c454794d3d7be2eadc894b4ed0a9801bae03d97a5760ae13b5c20059d2",
        "ResolvConfPath": "/var/lib/docker/containers/e637cbaba0f0bd0fc54fb656a56642dfbdd32e9c382a2a86a1b4d82fb372a5c5/resolv.conf",
        "HostnamePath": "/var/lib/docker/containers/e637cbaba0f0bd0fc54fb656a56642dfbdd32e9c382a2a86a1b4d82fb372a5c5/hostname",
        "HostsPath": "/var/lib/docker/containers/e637cbaba0f0bd0fc54fb656a56642dfbdd32e9c382a2a86a1b4d82fb372a5c5/hosts",
        "LogPath": "/var/lib/docker/containers/e637cbaba0f0bd0fc54fb656a56642dfbdd32e9c382a2a86a1b4d82fb372a5c5/e637cbaba0f0bd0fc54fb656a56642dfbdd32e9c382a2a86a1b4d82fb372a5c5-json.log",
        "Name": "/webserver",
        "RestartCount": 0,
        "Driver": "overlayfs",
        "Platform": "linux",
        "MountLabel": "",
        "ProcessLabel": "",
        "AppArmorProfile": "",
        "ExecIDs": null,
```



Conclusion:

In this experiment, application containerization was successfully demonstrated using Docker on a host system. The fundamental concepts of container technology were explored by searching, pulling, running, and managing Docker images and containers. The experiment verified how Docker provides lightweight, isolated, and highly portable environments capable of hosting operating systems like Ubuntu and application services like NGINX web server without the overhead of traditional virtual machines

Through hands-on execution, key skills were gained in navigating the container lifecycle, including searching registries, running interactive Ubuntu shells, configuring internal dependencies, and deploying background NGINX web services with host port mapping. Additionally, this experiment reinforced the practical distinction between static, immutable Docker images and active, mutable containers, highlighting how sharing the host OS kernel while maintaining absolute filesystem and process isolation ensures consistent runtime behavior across diverse host systems.